

Anthony Katona

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🔄 Summary Statement

Chemistry PhD specializing in polymer and ceramic coating processes, photothermal materials processing, and materials characterization. Experienced in lab-scale process development, class IV laser systems, safety oversight, and custom software and electronics development for productivity advancement. Skilled at communication to a wide range of knowledge levels and building processes that improve transparency and maintainability.

🎓 Experience & Education

The Pennsylvania State University

Chemistry (State College, PA)

2023–current

Postdoctoral Researcher

- 🔧 led laser-processing development for UHTC preceramic polymer coating precursors within an 8-group MURI collaboration, establishing experimental procedures, air-free workflows, and process-parameter screening methods
- 🔧 built remote monitoring instrumentation and a transparent flow/purge chamber for air-sensitive sample processing, improving real-time observation and control during laser treatment
- 🔧 designed nanosecond pyrometer to inform collaborating theory groups of accessible peak temperatures during laser pulses
- 🔧 integrated containerization into multiple existing project software pipelines with complex dependency stacks to facilitate access to tools between collaborators

The Pennsylvania State University

Chemistry (State College, PA)

2023

PhD

- 🔧 developed photo-thermal polymer techniques for coatings and 3D-printing applications
- 🔧 modeled and printed custom 3DP components for managing optics and improving in-house HFR videography and contact angle goniometry
- 🔧 synthesized ultra-fast/nano-scale thermal transport materials
- 🔧 developed Python analysis pipelines for the conversion of profilometry, mechanical testing, DMA, and thermal instrument output to reproducible metrics and plots
- 👤 oversaw lab safety as both General and Laser Safety Officer, including management of chemical inventory and coordinating laboratory audits
- 👤 managed supply and maintenance of pumps, gas cylinders, and gloveboxes
- 👤 taught as TA for all 5 available first-year chemistry courses for a total of 8 semesters
- 👤 co-developed and implemented novel thermal conductivity undergraduate lab experiments for *Dr. Bratoljub Milosavljevic's* physical chemistry course
- 👤 served as graduate student TA mentor for 1/3 of class and research mentor for 5 undergraduate researchers, teaching experimental design, instrument use, scripting for data analysis, and scientific writing

Susquehanna University

Biochemistry; Physics minor (Selinsgrove, PA)

2015

BS

- 🔧 synthesized transition metal Janus complexes with *Dr. William Dougherty* as a thesis project
- 🔧 assisted in galaxy classification model training with *Dr. Violet Mager* during summer research

Technical Skills

- 💡 **spectroscopy/microscopy:** NMR, IR, UV-Vis, AFM, TEM (STEM, EDS)
- 🏠 **materials:** mechanical (UTM, DMA) and thermal (DSC, TGA, TPS) tests; lathes, mills, 3DP, engravers
- ⚙️ **laser:** class IV CW/pulsed safety, handling, and optics
- 🔌 **electronics:** circuit design/prototyping, signal analysis, microcontrollers, LabVIEW
- 💻 **data/automation:** Python, Pandas/NumPy/SciPy, C#, C++, LabVIEW, HTML, CSS, JS, SQL, ImageJ, WEKA segmentation, Docker
- 🗣️ **technical communication:** LaTeX, markdown, Adobe CC (Ae, Il, Ps, Pt, Pr), Veusz, FL Studio, Blender 3D, Unreal Engine

Publications & Patents

publications:

- S Phillips *et al.*, 2026 review. *Indust. Chem. & Mat.*, **“Nanometer-Scale Surface Roughness...”**
- A Katona & B Lear, 2025. *J. Phys. Chem. C*, **“Photothermal Curing of Polydimethylsiloxane: Carbon Black Composites Results in Changes to Polymer Topography, Cross-Link Density, and Mass Density”**
- A Blasone *et al.*, 2025. **“Rapid Photothermal Curing of PDMS on Paper”**
- A Katona & B Lear, 2024. *Macromol.*, **“Effective photothermal curing of PDMS using ultra-low loadings of carbon black”**
- V Mager *et al.*, 2018. *ApJ* 864, **“Galaxy Structure in the Ultraviolet...”**

patent filings:

- Co-inventor, “Photothermal Generation of Ceramics,” U.S. Provisional Patent Application No. 63/910,435, filed Nov. 3, 2025
- Co-inventor, “Photothermal Curing of Polymers on Thermally Sensitive Substrates,” U.S. patent application pending; priority applications filed July, November, and December 2024

Outreach & Volunteer Work

Exploration-U

2019

- Presented a polymer science demonstration at Bellefonte High School.

Graduate Women in Science

2017–2018

- Presented a demonstration on quantum locking in superconductors for Girl Scouts.
- Designed and presented a demonstration on the physics of sound for Girl Scouts.
- Designed and presented a demonstration on the psychology of bias for Girl Scouts.

Pennsylvania Junior Academy of Science

2017

- Served as a judge for student science presentations at the Penn State regional competition.